

INTERNSHIP REPORT

**FIBER OPTIC
COMMUNICATIONS,
FTTH NETWORK ARCHITECTURE
AND ELECTRICAL HOME
APPLIANCES
MAINTENANCE**



Submitted by:

RIFA

BSc Physics Student

Under the guidance of:

Dr. Mohemmed Shanid N A

Department of Physics

Sullamussalam Science College, Areekode

CERTIFICATE

This is to certify that the internship report entitled “Fiber Optic Communications, FTTH Network Architecture and Electrical Home Appliances Maintenance” is a bona fide record of the industrial training carried out by **RIFA** under my academic supervision and guidance, in corporate collaboration with Bfone Fibernet, Edavannappara, in partial fulfillment of the require-

Dr. Mohemmed Shanid N A

ments for the award of the Degree of Bachelor of Science in Physics. Head of the Department

Department of Physics

Internal Examiner

Contents

INTERNSHIP ACTIVITY SCHEDULE	ii
I Fiber Optic Communications and FTTH Network Architecture	1
Part Introduction: Fiber Optic Infrastructure	2
1 Introduction to Optical Fiber Communication	3
1.1 Overview	3
1.2 The Basic Fiber Optic Link	3
1.3 Advantages of Fiber Optics	3
2 Principle and Cable Structure	5
2.1 Principle of Operation: Total Internal Reflection	5
2.2 Cable Anatomy	5
2.3 Cable Types	5
3 Link Budgeting, Splicing, and Mathematical Calculations	7
3.1 OFC Splicing	7
3.2 Logarithmic Conversion Mechanics	7
3.3 Splitter Loss Calculations	7
3.4 Standard Loss Values for Link Performance	7
3.5 Mathematical Verification: Power Link Budget Case Study	8
3.5.1 Case Study Parameter Blueprint	8
3.5.2 Link Status Analysis	8
4 Optical Fiber Splicing: Detailed Analysis	10
4.1 What is Splicing?	10
4.2 Types of Splicing	10
4.2.1 Fusion Splicing	10
4.2.2 Mechanical Splicing	10
4.3 Importance and Significance	10
4.4 Step-by-Step Fusion Splicing Process	11
4.5 Advantages	11
4.6 Future Scope	11
4.7 Conclusion	11

5	Introduction to FTTH & Passive Optical Networks (PON)	12
5.1	Overview of FTTH	12
5.2	The FTTH Network Architecture Layers	12
5.3	Core Hardware Components	12
5.3.1	Active Components (Power Required)	13
5.3.2	Passive Components (No Power Required)	13
6	FTTH Deployment, Coding and Testing	14
6.1	Fiber Color Coding Standard (TIA-598-C)	14
6.2	Physical Deployment & Site Survey	14
6.2.1	Site Survey & Planning	14
6.2.2	Outdoor Routing Operations	15
6.2.3	Indoor Termination	15
6.3	Splicing Methods Comparison	15
6.4	Link Budget & Testing Benchmarks	15
6.4.1	Wavelength Multiplexing	15
6.4.2	Testing Parameters	15
6.5	Troubleshooting & Indicators	16
7	Field Visit: BSNL Exchange Office, Manjeri	17
7.1	Introduction to the Field Visit	17
7.2	Observations and Core Infrastructure	17
7.3	Conclusion of the Visit	17
	Part Conclusion: Fiber Optic Infrastructure	18
II	Electrical Home Appliances Maintenance and Troubleshooting	19
	Part Introduction: Domestic Electrical Engineering	20
8	Fundamentals of Electrical Systems & Safety	21
8.1	Introduction	21
8.2	Fundamentals of Electrical Systems & Safety Overview	21
8.2.1	Circuit Behaviors & Laws	21
8.2.2	Earthing and Protection Systems	21
9	Technical Breakdown & Diagnostics of Core Home Appliances	23
9.1	Electric Iron	23
9.2	Induction Cooker	24
9.3	Mixer Grinder	24
9.3.1	Ceiling Fan	25

9.3.2 Wall Fan	25
10 Troubleshooting & Maintenance Practice Matrix	27
10.1 Internship Outcomes & Skills Acquired	28
11 Applied Practical Calculations	30
11.1 Domestic Electricity Consumption Auditing	30
11.2 Charge Calculation in Audio Systems	30
Part Conclusion: Domestic Electrical Engineering	31

Internship Activity Schedule

The day-wise internship activities and topics covered during the training period are summarized in the table below:

Day	Date	Topics / Activities Covered
Day 1	21.05.2026	Optical Fiber Theory Class (Introduction to OFC)
Day 2	22.05.2026	Splicing Practical Section
Day 3	23.05.2026	Repairing of the Ceiling Fan
Day 4	25.05.2026	• Theory Class of Fiber To The Home (FTTH) • Repairing of Mixer Grinder • Field Visit of Splicing
Day 5	26.05.2026	Field Visit at Manjeri Telephone Exchange
Day 6	08.06.2026	Power Calculation Theory Class
Day 7	09.06.2026	• Repairing of Iron Box • Repairing of Wall Fan • Repairing of Induction Cooker

Table 1: Day-wise Internship Activity Log

Part I

Fiber Optic Communications and FTTH Network Architecture

Part Introduction: Fiber Optic Infrastructure

The modern telecommunication landscape is fundamentally powered by high-speed data transmission systems that rely on optical physics. Moving beyond legacy copper-wire networks, optical fibers guide light signals to transport large amounts of data over vast distances with minimal signal degradation. This initial module presents a comprehensive technical study of contemporary Fiber-Optic Communication (OFC) and Fiber-to-the-Home (FTTH) infrastructures, stemming from training at Bfone Fibernet. The core purpose here is to outline the entire journey of the light signal, from the hardware source down to the home ONT. The chapters in this section study Total Internal Reflection (TIR) guiding light in glass cores, multi-core cable design, and the calculations needed for fiber link budgets. Moreover, we document standard field procedures, including precision splicing, color code identification, and troubleshooting techniques used by technicians. This comprehensive study concludes with a field analysis of a regional telecommunication distribution center at the BSNL Exchange Office.

Chapter 1

Introduction to Optical Fiber Communication

1.1 Overview

Optical Fiber Communication (OFC) is a method of transmitting information from one place to another by sending pulses of light through an optical fiber. In this system, light acts as an electromagnetic carrier wave modulated to carry digital and analog data.

1.2 The Basic Fiber Optic Link

A typical optical fiber communication link consists of several key components that facilitate the transmission of data:

- Encoder/Source: Takes input data and translates it into digital electrical pulses.
- Transmitter (E/O): Uses a semiconductor LED or laser diode to convert electrical pulses into light signals.
- Optical Fiber: The silica glass or plastic medium that guides the light pulse over a distance.
- Receiver (O/E): Employs a photodiode (PIN or APD) to detect the light signal and convert it back to electrical form.
- Decoder: Reconstructs the original digital data from the electrical pulses.

1.3 Advantages of Fiber Optics

- Non-conductive: Made of glass/plastic, eliminating electrical spark risks and grounding problems.
- Electromagnetic Immunity: Unaffected by electromagnetic interference (EMI), radio frequency interference, or lightning strikes.
- Large Bandwidth: Capable of supporting incredibly high data rates and channel capacities.

- Low Loss: Extremely low signal attenuation, allowing long spans without repeaters.
- Small and Lightweight: Thin profile and lightweight nature simplify installation in conduits and aerial runs.

Chapter 2

Principle and Cable Structure

2.1 Principle of Operation: Total Internal Reflection

Light travels through the fiber core based on the principle of Total Internal Reflection (TIR). The occurrence of TIR requires two strict conditions:

- The core material must have a higher refractive index (n_1) than the surrounding cladding (n_2).
- The light must strike the core-cladding boundary at an angle of incidence exceeding the critical angle (θ_c).

The critical angle is calculated as:

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) \quad (2.1)$$

2.2 Cable Anatomy

A standard OFC cable consists of several protective and structural layers:

- Core: The central glass or plastic channel through which the light pulses travel.
- Cladding: The surrounding layer of glass with a lower refractive index that keeps the light trapped in the core.
- Buffer Coating: A plastic layer surrounding the cladding to shield it from moisture and mechanical stress.
- Strength Member: High-tensile aramid yarn (Kevlar) or steel wire to absorb tension during pulling.
- Outer Jacket: The tough plastic outer sheath (PVC or PE) protecting the cable from environmental factors.

2.3 Cable Types

- Tight-Buffer Cable: Features thick buffer coating directly on the fiber, ideal for indoor routing and patch cords.

- Loose-Tube Cable: Fibers lie loose inside gel-filled tubes, providing excellent protection against water and temperature shifts in outdoor ducts.
- Ribbon Cable: Multiple fibers aligned in flat rows, maximizing fiber density in tight spaces.

Chapter 3

Link Budgeting, Splicing, and Mathematical Calculations

3.1 OFC Splicing

Splices are permanent connections between two fibers. The main methods are:

- Fusion Splicing: Fusing the two glass ends together using a high-voltage electric arc.
- Mechanical Splicing: Aligning fiber cores within a precision alignment fixture and securing them.

3.2 Logarithmic Conversion Mechanics

Optical power values are measured using the logarithmic decibel-milliwatt (dBm) scale to make it easy to calculate losses across long fiber links:

$$P_{\text{dBm}} = 10 \log_{10} \left(\frac{P_{\text{mW}}}{1 \text{ mW}} \right) \quad (3.1)$$

3.3 Splitter Loss Calculations

The theoretical insertion loss of an optical splitter scales based on the number of output ports (N):

$$\text{Loss}_{\text{Ideal}} = 10 \log_{10}(N) \quad (3.2)$$

3.4 Standard Loss Values for Link Performance

To perform reliable link calculations, the engineering team uses the following standard attenuation parameters:

Network Element Component	Standard Loss Value
Fiber Path Loss @ 1310 nm / 1490 nm	0.40 dB per km
Fiber Path Loss @ 1550 nm	0.25 dB per km
Fusion Splice Junction	0.10 dB per splice
SC/UPC Physical Connector Pair	0.35 dB per connector
SC/APC Physical Connector Pair	0.25 dB per connector
1 × 2 Splitter Module	3.0 dB average insertion loss
1 × 16 Splitter Module	13.0 dB average insertion loss
1 × 32 Splitter Module	16.0 dB average insertion loss

Table 3.1: Standard Loss Parameters Used in FTTH Networks

3.5 Mathematical Verification: Power Link Budget Case Study

$$P_r = P_t - \text{Loss}_{\text{total}} \quad (3.3)$$

3.5.1 Case Study Parameter Blueprint

Consider a regional fiber run managed by Bfone Fibernet spanning a total distance of 7 km. The link includes 4 fusion splices, 2 SC/UPC connector pairs, a 1 × 16 splitter, and a 1 × 8 cascading splitter module. The loss budget details for this specific fiber span are calculated below:

- Fiber line loss: 7 km of fiber at 0.40 dB/km results in a loss of 2.8 dB
- First splitter stage (1 × 16) loss: 13.0 dB
- Second splitter stage (1 × 8) loss: 10.0 dB
- Fusion splice losses: 4 splices at 0.10 dB each equals 0.4 dB
- Connector losses: 2 SC/UPC pairs at 0.35 dB each equals 0.7 dB

Summing these individual loss values yields the total attenuation for the link:

$$\begin{aligned} \text{Total Attenuation Loss} &= 2.8 + 13.0 + 10.0 + 0.4 + 0.7 \\ &= 26.9 \text{ dB} \end{aligned}$$

3.5.2 Link Status Analysis

Standard ONT modems require an input signal level within an optical power range between -10 dBm and -28 dBm to operate reliably: Minimum Receiver Threshold $\geq$ -28 dBm

$$\text{Receiver Sensitivity Threshold} \leq -28 \text{ dBm} \quad (3.4)$$

Because the calculated total loss for this link is 26.9 dB, a typical transmitter power output of 0 dBm ensures that the received power level stays above the receiver's minimum sensitivity threshold (-26.9 dBm $\geq$ -28 dBm). This mathematical verification confirms that the fiber link will function correctly and maintain high signal reliability.

Chapter 4

Optical Fiber Splicing: Detailed Analysis

4.1 What is Splicing?

Splicing is the process of joining two optical fiber cores permanently to create a continuous optical path. While connectors are built for quick disconnects, splices are permanent joints optimized for stability. The primary design target is to keep optical attenuation as low as possible and prevent back-reflection.

4.2 Types of Splicing

The main methods used in the industry are:

4.2.1 Fusion Splicing

This is the most reliable and industry-standard method. It involves using an electric arc to melt and fuse two fiber ends together. Attenuation: Usually between 0.01 dB and 0.05 dB. Key Applications: Long-haul networks, fiber backbones, and FTTH distribution.

4.2.2 Mechanical Splicing

This method uses a mechanical fixture to align the fiber ends and an index-matching gel to facilitate light transmission between the cores. Attenuation: Usually ranges from 0.2 dB to 0.5 dB. Key Applications: Quick field repairs, testing setups, and situations without access to fusion splicers.

4.3 Importance and Significance

- **Link Extension:** Joins shorter fiber spools to build long-distance network runs.
- **Signal Integrity:** Assures optimal light propagation and very low insertion loss.
- **Back-reflection Control:** Limits return loss (reflections) to protect high-power laser transmitters.
- **Durability:** Creates robust, long-lasting joints that resist temperature and physical strain.

4.4 Step-by-Step Fusion Splicing Process

1. Stripping: Stripping away the 250 μm polymer buffer coating using precision fiber strippers to expose bare glass.
2. Cleaning: Cleaning the bare glass surface with 99% pure Isopropyl Alcohol (IPA) and lint-free wipes.
3. Cleaving: Cutting the fiber using a precision cleaver to achieve a flat 90-degree angle (less than 0.5 degrees deviation).
4. Fusion: Aligning the fibers automatically using PAS and applying a localized electric arc to fuse the glass at 2000°C.
5. Protection: Sliding a heat-shrink protector sleeve over the splice point and baking it in the splicer's oven to seal it.

4.5 Advantages

- Provides the absolute lowest signal attenuation (≤ 0.05 dB).
- Forms a highly stable, permanent glass-to-glass bond.
- Unbothered by mechanical vibrations and thermal expansion.
- Very compact, fitting easily into splice trays and closures.

4.6 Future Scope

- Automation: Introduction of automated splicing machines and robotics for harsh environments.
- Multi-core Fiber Splicing: Upgrading equipment to splice multi-core fibers that support higher spatial bandwidth.
- Enhanced Precision: Incorporating nanometer-level alignment to reach near-zero signal loss in critical paths.

4.7 Conclusion

Optical fiber splicing is the foundation of modern high-speed communication. By ensuring a seamless and permanent joint, it guarantees that data travels across vast distances with negligible loss. While fusion splicing is the preferred method for permanent infrastructure, mechanical splicing remains a vital tool for rapid, field-based repairs.

Chapter 5

Introduction to FTTH & Passive Optical Networks (PON)

5.1 Overview of FTTH

Fiber-to-the-Home (FTTH) represents the ultimate evolutionary step in broadband access networks, replacing traditional copper-based infrastructures (like DSL or Coaxial cables) entirely with optical fiber lines directly to individual subscriber premises. The dominant technology in FTTH is PON, or Passive Optical Network. It is 'passive' because there are no powered switches in the field; light is divided using glass splitters. This design reduces power consumption, cuts down on hardware failures, and lowers operating costs. Through this point-to-multipoint topology, a single fiber feeds multiple households.

5.2 The FTTH Network Architecture Layers

The physical infrastructure of an FTTH network is divided into three distinct geographical segments:

- A. The Feeder Segment: Connects the Central Office (CO) to the Fiber Distribution Hub (FDH). It uses heavy, high-fiber-count cables (48 to 144+ cores) to carry signals over long trunk routes (up to 20 km) without splitting.

- B. The Distribution Segment: Spans from the FDH to street-level boxes like the Optical Distribution Point (ODP). Feeder fibers are split at the FDH, and distribution cables (12 to 24 cores) run along streets on poles or in ducts.

- C. The Drop Segment: The final stretch running from the ODP to the customer's ONT. It uses a single or dual-core drop cable designed with high bend tolerance (G.657 standard) to navigate tight corners inside home conduits.

5.3 Core Hardware Components

An FTTH network relies on a strict handshake between active hardware (requiring power) and passive hardware (no power):

5.3.1 Active Components (Power Required)

OLT (Optical Line Terminal): The central switch situated at the Central Office. It translates backbone internet traffic into optical pulses and coordinates downstream/upstream transmission scheduling.

ONT / ONU (Optical Network Terminal / Unit): The subscriber-end modem. It converts the incoming light signals from the drop cable into standard electrical Ethernet or Wi-Fi signals for household use.

5.3.2 Passive Components (No Power Required)

Optical Splitters: Passive glass components that divide a single input light wave into multiple output paths (e.g., 1:8, 1:16, 1:32) to feed multiple subscribers without power.

FDH / ODF Cabinets: Weatherproof enclosures hosting splitters and splice trays, securing connections against dust, water, and heat.

Chapter 6

FTTH Deployment, Coding and Testing

6.1 Fiber Color Coding Standard (TIA-598-C)

To safely identify individual fibers inside a multi-core cable, technicians use the industry-standard 12-color coding sequence. For cables with more than 12 fibers, the sequence repeats with a physical black tracer/stripe.

No.	Fiber Color	Symbol	No.	Fiber Color	Symbol
1	Blue	BL	7	Red	RD
2	Orange	OR	8	Black	BK
3	Green	GR	9	Yellow	YL
4	Brown	BR	10	Violet	VI
5	Slate (Grey)	SL	11	Rose (Pink)	RS
6	White	WH	12	Aqua	AQ

Table 6.1: TIA-598-C Color Coding Identification

6.2 Physical Deployment & Site Survey

6.2.1 Site Survey & Planning

Before installation, technicians perform a comprehensive site survey consisting of several phases:

- **Route Mapping:** Planning the shortest cable path from the street ODP to the customer premises.
- **Structural Assessment:** Finding entry points and deciding on the best indoor ONT location.
- **Safety Check:** Checking for power lines to maintain safe spacing and avoid electrical hazards.
- **BOM Calculation:** Creating the Bill of Materials including clamps, drop cables, and wall outlets.

6.2.2 Outdoor Routing Operations

Aerial Deployment: Drop wires are suspended between poles using drop clamps. Technicians leave slack loops to accommodate expansion and contraction from temperature changes.

Underground Deployment: In suburban areas, fiber is pulled through buried protective HDPE ducts to shield it from excavation and weather damage.

6.2.3 Indoor Termination

The fiber cable enters the home and is terminated at a wall-mounted fiber outlet or rosette box. A bend-insensitive patch cord (G.657 standard) is then run from the outlet to the ONT, allowing tight bends around walls without causing signal loss.

6.3 Splicing Methods Comparison

Feature	Fusion Splicing	Mechanical / Fast Connectors
Mechanism	Uses a high-voltage electric arc to melt and fuse glass tips permanently.	Physically aligns two fiber ends inside a sleeve using index-matching gel.
Signal Loss	Extremely low (< 0.05 dB to 0.1 dB).	Slightly higher (0.2 dB to 0.5 dB).
Pros & Cons	Most reliable, but requires an expensive fusion splicer machine.	Fast setup, low initial tool cost. Ideal for home drop termination.

Table 6.2: Comparison of Fusion and Mechanical Splicing

6.4 Link Budget & Testing Benchmarks

FTTH operates over single-mode fiber using separate bidirectional wavelengths to avoid interference:

6.4.1 Wavelength Multiplexing

- 1490 nm: Carries downstream internet and voice traffic from OLT to subscriber ONT.
- 1310 nm: Used for upstream traffic uploads from ONT to the central OLT.
- 1550 nm: Allocated for downstream analog or digital TV overlays.

6.4.2 Testing Parameters

- Optical Power Meter (OPM): Used to measure the optical power level at the ONT. The normal range is -15 dBm to -25 dBm. Levels below -27 dBm indicate an attenuation fault.
- OTDR (Optical Time Domain Reflectometer): Sends light pulses down the fiber to analyze backscatter, mapping splice points and finding the exact location of cable cuts.

6.5 Troubleshooting & Indicators

Field diagnostics rely heavily on the ONT hardware status LED combinations:

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 6.3: ONT Troubleshooting Matrix

Chapter 7

Field Visit: BSNL Exchange Office, Manjeri

7.1 Introduction to the Field Visit

As part of the internship in optical fiber communication, a technical visit was conducted to the BSNL Exchange office, Manjeri to gain practical exposure to the operation and management of fiber-optic communication networks. This excursion let us trace the physical flow of data from central carrier routers to the local distribution loops.

7.2 Observations and Core Infrastructure

During the visit, we observed different equipment used in the exchange, including the Optical Line Terminal (OLT), fiber distribution panels, and other networking devices. The officials explained how internet, voice, and other communication services are transmitted through optical fiber cables from the exchange to customers. The OLT works in tandem with key backend routing and authentication nodes: A Broadband Network Gateway (BNG) handles routing, authenticates user credentials, and controls access policies. The MNGN platform provides carrier-grade packet transport, integrating FTTH broadband, IPTV, and business data links onto a unified network. The OLT connects to customer ONTs via the optical distribution network.

7.3 Conclusion of the Visit

The visit provided practical knowledge about the working of a telecommunication exchange and helped us relate the theoretical concepts of optical fiber communication to real-world applications. It was a valuable learning experience that improved our understanding of modern fiber-optic communication systems.

Part Conclusion: Fiber Optic Infrastructure

In conclusion, this module has successfully detailed the comprehensive operational framework of modern high-speed telecommunications. We studied light propagation in single-mode cores and analyzed fiber structure and attenuation in distribution cables. Calculations for splitters and link budgets demonstrated how signal integrity is maintained over a passive network. Practical exercises at Bfone Fibernet allowed us to master fusion splicing, color code identification, and OTDR troubleshooting. Additionally, the BSNL Exchange visit illustrated macro-scale network architecture, showing the interaction between OLTs, BNG routers, and ONTs. Ultimately, this section showed that reliable gigabit broadband requires both accurate math and high-quality field installation.

Part II

Electrical Home Appliances Maintenance and Troubleshooting

Part Introduction: Domestic Electrical Engineering

The integration of specialized electro-mechanical hardware remains a vital pillar of the modern domestic environment. Devices utilizing resistive heating, induction cooking, or single-phase induction motors apply core principles of thermodynamics and electromagnetism. This second part focuses on the electrical systems, diagnostics, and repair procedures learned during the hands-on sessions. The goal was to build a solid diagnostic mindset based on electrical theory rather than basic parts-swapping. The following chapters cover domestic wiring, series and parallel circuits, and safety devices like grounding, MCBs, and RCCBs that protect against shocks and short circuits. We then detail the construction and troubleshooting of seven appliances: the electric iron, induction cooker, mixer grinder, water geyser, ceiling fan, wall fan, and utility pump. This foundational mechanical study finishes with applied practical audits covering consumer load accounting and electrical charge tracking.

Chapter 8

Fundamentals of Electrical Systems & Safety

8.1 Introduction

The modern domestic environment relies entirely on the seamless operation of electrical appliances. From food preparation to climate control and water management, these devices have transitioned from luxuries to everyday essentials. However, continuous operation inevitably leads to mechanical wear, insulation degradation, and electrical faults. This module was completed to apply electrical theory to practical repair. The course covered safety protocols, wiring layouts, and the step-by-step repair of seven core home appliances. The focus was on understanding operational physics—such as Joule heating, magnetic fields, and fluid flow—while strictly following safety rules.

8.2 Fundamentals of Electrical Systems & Safety Overview

Before working on any appliance, a solid foundation in circuit behavior and safety infrastructure was established.

8.2.1 Circuit Behaviors & Laws

- Ohm's Law & Power: We use $V = IR$ and $P = VI$ to calculate normal resistance values, allowing us to find short circuits and open windings during testing.
- Series Wiring: Current is constant while voltage drops. Used for safety switches, fuses, and thermostats to interrupt current flow.
- Parallel Wiring: Voltage remains constant while current divides. Standard for home wiring to ensure appliances operate independently.

$$V = IR \quad \text{and} \quad P = VI = I^2R \quad (8.1)$$

8.2.2 Earthing and Protection Systems

- Earthing: Connects the metal body of an appliance to the ground. If a hot wire touches the chassis, current flows to earth, preventing shock to the user.

- Fuses & MCBs: Devices that open the circuit during short circuits or overloads, protecting wiring from overheating.
- RCCB / ELCB: Detects small leakages (30mA) to ground by comparing live and neutral currents, tripping instantly to prevent severe shock.

Chapter 9

Technical Breakdown & Diagnostics of Core Home Appliances

9.1 Electric Iron

A. Working Principle

An electric iron works on the principle of the Joule's Heating Effect of Electric Current. When an electric current passes through a high-resistance conductor (heating element), it generates thermal energy ($H = I^2 R t$). A bimetallic strip thermostat controls the heat. Since the two bonded metals expand at different rates, heating bends the strip to open contacts and cut the circuit. Cooling resets the strip, restarting the heating cycle.

$$H = I^2 R t \quad (9.1)$$

B. Key Components and Parts

- Soleplate: A smooth, heavy metal plate (aluminum/cast iron) that transfers heat to the fabric.
- Heating Element: A nichrome wire ribbon wrapped in mica sheets to insulate the live voltage while conducting heat.
- Bimetallic Thermostat: The temperature regulator with an adjustable dial that sets the contact gap.
- Thermal Fuse: A safety fuse that melts permanently if temperatures exceed safe limits due to thermostat failure.

C. Appliance Conclusion

Through the study of the electric iron, the practical application of electrothermal conversion and thermal feedback loops via bimetallic components was fully understood. Testing this device highlighted how crucial thermal fuses are in preventing catastrophic overheating.

9.2 Induction Cooker

A. Working Principle

Unlike traditional cooking methods that use thermal conduction, an induction cooker operates on the principles of Electromagnetism and Eddy Currents. An internal inverter converts 50Hz AC mains into high-frequency AC (20-40kHz). This current flows through an induction coil to generate an alternating magnetic field. Placing a ferromagnetic pan on the glass induces eddy currents in the pan's base. The pan's electrical resistance causes it to heat up directly, leaving the glass top relatively cool. **B. Key Components and Parts**

- Induction Coil: A flat, spiral copper coil that creates the alternating magnetic field.
- IGBT: A high-power semiconductor switch that drives the high-frequency inverter circuit.
- Main PCB & Control Unit: The controller that processes settings, manages power level switching, and monitors safety sensors.
- Fan & Temperature Sensor (RTD): A cooling fan and thermistor that prevent overheating and detect boil-dry conditions.

C. Appliance Conclusion

The induction cooker diagnostics provided essential insights into solid-state electronics and high-frequency power switching. It demonstrated how magnetic fields can be manipulated for localized heating, emphasizing the need for robust PCB-level troubleshooting.

9.3 Mixer Grinder

A. Working Principle

The mixer grinder operates on electromagnetic mechanical conversion using an AC Universal Motor. A universal motor is a series-wound motor designed to run on either AC or DC power. When voltage is applied, current flows through both the stator (field windings) and the rotor (armature windings) via carbon brushes. This creates opposing magnetic fields, producing a high starting torque and high rotational speeds (up to 20,000 RPM), which are necessary for chopping, blending, and grinding food materials. **B. Key Components and Parts**

- Stator & Armature: The stationary field coils and rotating core that convert electrical energy to torque.
- Commutator & Brushes: Sliding carbon contacts that deliver current to the rotating commutator bars.

9.3.1 Ceiling Fan

A. Working Principle

A ceiling fan operates on the principle of a Single-Phase Induction Motor (Permanent Split Capacitor Motor). Single-phase AC power cannot generate a rotating magnetic field on its own to start a motor. To solve this, the motor has two separate windings: a Starting Winding and a Running Winding. A capacitor is wired in series with the starting winding to create a 90° electrical phase shift in the current. This phase shift creates a simulated rotating magnetic field, which induces a current in the squirrel-cage rotor, generating the torque needed to turn the fan blades. **B. Key Components and Parts**

- Stator & Rotor: An inner stationary core with copper windings and an outer rotating steel shell.
- Run Capacitor: A 2.25 to 2.5 microfarad capacitor that creates the phase shift to start the motor.
- Ball Bearings: Dual bearings that support the rotor shell, minimizing noise and friction.
- Down-Rod: The steel pipe assembly that suspends the fan safely from the ceiling hanger.

C. Appliance Conclusion

Working with ceiling fans clarified the theory of phase splitting in single-phase induction systems. The practical experience emphasized the diagnostic process of isolating electrical winding failures from mechanical bearing resistance.

9.3.2 Wall Fan

A. Working Principle

A wall fan operates similarly to a ceiling fan, using a Single-Phase Permanent Split Capacitor Induction Motor, but includes specialized mechanisms for targeted directional airflow and oscillation. While the motor creates mechanical rotation using a starting winding, running winding, and a capacitor to shift the current phase, it also drives a mechanical synchronous link gear system or an independent secondary micro-motor to achieve horizontal oscillation (swiveling back and forth). Speed regulation is managed either via a multi-tapped transformer winding or an electronic TRIAC circuit on a PCB, controlling voltage delivery to the motor. **B.**

Key Components and Parts

- Blades & Grille: Curved plastic/metal blades protected by a wire mesh safety guard.
- Oscillation Gearbox: A reduction gear assembly that converts high-speed motor rotation into a slow swiveling motion.

- Speed Selector: A mechanical rotary switch, pull-cord, or electronic circuit that adjusts voltage to regulate speed.
- Thermal Cut-Off (TCO): A thermal fuse inside the stator coils that cuts power if the motor overheats due to a jam.

C. Appliance Conclusion

The analysis of the wall fan highlighted how structural mechanisms (linkage gears) integrate with electrical induction motors. It offered an excellent case study on thermal protection embedded within motor windings and the diagnostic tracking of mechanical oscillation failures.

Chapter 10

Troubleshooting & Maintenance Practice Matrix

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 10.1: ONT Troubleshooting Matrix Summary

Appliance	Common Symptom	Root Cause Detected	Corrective Action Performed
Electric Iron	No heating; indicator light off	Open circuit in thermal fuse / Broken power cord	Replaced thermal fuse; trimmed and re-terminated power cord.
Induction Cooker	Dead display; won't power up	Blown main fuse due to shorted IGBT	Replaced the shorted IGBT and main glass fuse; checked gate driver circuit.
Mixer Grinder	Sparks from bottom; sudden stop	Worn carbon brushes / Overload switch tripped	Replaced carbon brushes; cleared jar jam and reset the bottom OLP switch.
Water Heater	Water not heating; indicator on	Scale buildup or open-circuit in heating element	Drained tank, removed limescale, and replaced the immersion heating element.
Ceiling Fan	Humming sound; spins very slowly	Degraded or leaked capacitor	Desoldered/disconnected old capacitor, verified value, and installed new 2.5 μ F unit.
Wall Fan	Fan rotates but does not oscillate / Overheating	Stripped linkage gears / Dry rotor shaft	Replaced the plastic gear wheel assembly; cleaned and lubricated the rotor shaft with machine oil.
Pump Motor	Motor hums but does not rotate	Seized bearings / Defective starter capacitor	Freed impeller manually; replaced bearings and capacitor.

Table 10.2: Appliance Troubleshooting & Maintenance Practice Matrix

10.1 Internship Outcomes & Skills Acquired

- **Safety First Mentality:** Learned to always check for live voltage using a test pen, work on rubber mats, and verify ground lines before powering up devices.
- **Structured Troubleshooting:** Mastered the process of tracing faults along the circuit path, starting from the plug, checking fuses, and then testing switches and windings.

- Multimeter Proficiency: Gained hands-on experience using multimeters to test continuity, winding resistances, and checking live AC voltages safely.

Chapter 11

Applied Practical Calculations

11.1 Domestic Electricity Consumption Auditing

To calculate household power consumption, energy is tracked in kilowatt-hours (kWh), where 1 Unit = 1 kWh.

Determine the exact 30-day billable consumption for a household running two 40W tube lights for 6 hours daily, three 65W fans for 8 hours daily, and separate appliances drawing 150W for 2 hours daily.

$$E_{\text{Lights}} = 2 \times 40 \text{ W} \times 6 \text{ hrs/day} \times 30 \text{ days} = 14,400 \text{ Wh} = 14.4 \text{ kWh}$$

$$E_{\text{Fans}} = 3 \times 65 \text{ W} \times 8 \text{ hrs/day} \times 30 \text{ days} = 46,800 \text{ Wh} = 46.8 \text{ kWh}$$

$$E_{\text{Misc}} = 150 \text{ W} \times 2 \text{ hrs/day} \times 30 \text{ days} = 9,000 \text{ Wh} = 9.0 \text{ kWh}$$

$$E_{\text{Total}} = 14.4 \text{ kWh} + 46.8 \text{ kWh} + 9.0 \text{ kWh} = 70.2 \text{ kWh (Units)}$$

11.2 Charge Calculation in Audio Systems

A 230V system consumes a total of 8 MJ of energy during operation. Determine the absolute electrical charge that moved through the circuit.

$$\text{Given: } V = 230 \text{ V, } E = 8 \times 10^6 \text{ J}$$

$$\text{Equation: } E = qV$$

$$\begin{aligned} q &= \frac{E}{V} \\ &= \frac{8 \times 10^6 \text{ J}}{230 \text{ V}} \\ &= 3.48 \times 10^4 \text{ Coulombs (C)} \end{aligned}$$

Part Conclusion: Domestic Electrical Engineering

In conclusion, this second module has established a robust technical bridge between undergraduate physics and field-level electrical repair logic. Studying the components and wiring of seven core appliances provided valuable insight into domestic electrical layouts. Checking electrical behaviors, like thermostat switching in irons or capacitor values in induction motors, showed the value of multimeter testing. The exercises emphasized the importance of safety devices like thermal cut-offs (TCOs) and overload protectors (OLPs) in preventing damage. Furthermore, calculating energy consumption and charge transfer helped us understand power management. Ultimately, the hands-on repair skills and diagnostic logic learned during this course form a valuable toolkit for electrical maintenance.